

Blessing O. Okosun

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Education

Doctor of Philosophy - Biology <i>University of North Dakota, Grand Forks, ND</i> Advisor: Diane Darland, Ph.D. Dissertation: <i>Vascular-derived factors and their influence on the developing mouse cortex</i>	<i>Dec. 2026</i>
Communicating Science Graduate Certificate <i>University of North Dakota, Grand Forks, ND</i>	<i>May 2025</i>
Micro certification - Mentors Helping Mentors <i>University of North Dakota, Grand Forks, ND</i>	<i>Feb. 2024</i>
Bachelor of Science in Biology <i>Dickinson State University, Dickinson, ND</i> Major: <i>Cell and Molecular Biology</i> Minor: <i>Chemistry</i>	<i>May 2021</i>
Bachelor of Science in Science Laboratory Technology <i>University of Benin, Benin City, Nigeria</i> Major: <i>Microbiology</i>	<i>Dec. 2015</i>
Nigerian Certificate of Education <i>College of Education, Ekiadolor, Nigeria</i>	<i>March 2009</i>

Skills

Molecular & Cellular Biology <i>Flow cytometry, Immunolabeling, gel electrophoresis, PCR, RT-qPCR, DNA/RNA extraction, Molecular sub-cloning, Mammalian (stem) cell culture, Co-culture and organoid models</i>
Bioinformatics <i>Phylogenetic analysis, single-cell & bulk RNA-Seq analysis</i>
Advanced Imaging <i>Specialized in the use of different confocal microscopy, methodologies, quantitative microscopy workflows and imaging analysis</i>

Research Experience

Graduate Research Assistant <i>University of North Dakota, Grand Forks, ND</i> Supervisor: <i>Dr. Diane Darland</i>	<i>Summer Semester 2021 - Present</i>
• 2025; Using flow cytometric analysis, immunolabeling, single-cell analysis, and confocal microscopy to determine the role of leptin/leptin receptors in determining neural stem cell fate. I also worked with a collaborator on the bioapplication of nanozymes for detecting hydrogen peroxide release from metabolically stressed endothelial cells.	

- 2024; Used phylogenetic analysis to determine the evolutionary relationship of leptin receptors across different members of Deuterostomes. I also worked with a collaborator on the bioapplication of nanozymes for the detection of reactive oxygen species in metabolically stressed endothelial cells.
- 2022 - 2023; I utilized co-culture and organoid models, along with bulk RNA sequencing, PCR, and RT-qPCR techniques, to investigate how the differential distribution of vascular endothelial growth factor isoforms affects the expansion of the embryonic mouse cortex. Additionally, I focused on the bioapplication of polymer-based nanoparticles carrying siRNA to knock down the expression of selected genes in brain-derived microvascular endothelial cells and fibroblasts. Furthermore, I explored the bioapplication of persistent luminescence polymer-based nanoparticles in brain-derived microvascular cells.
- 2021; used magnetic beads and antibodies to select for neural stem cells from mice embryos' brains and generate clonal cell lines.

IDeA Networks of Biomedical Research Excellence Research Assistant

Sept. 2019 - Jan. 2021

Dickinson State University, Dickinson, ND

Supervisor: Dr. Craig Whippo

- Determine different Arabidopsis responsiveness to volatile organic compounds. This was done by characterization of the phenotypic natural variations of the different parts of imaged Arabidopsis that were grown in media. Techniques employed were gas sterilization, image analysis on ImageJ, and data processing in R.
- Carried out a Genome wide Association Study to determine the genetic variation and diversity of 198 different Arabidopsis accessions.

NSF Research Experiences for Undergraduates

May. 2019 - Aug. 2021

South Dakota School of Mines & Technology

Supervisor: Dr. Timothy Brenza & Jordan A. Hoops

- Characterization of Calu-3 cell line culture protocol by determining if air-liquid interface method (ALI; that mimics cell conditions in the lungs) is better for drug delivery compared to the traditional method of submerging cells in media.
- Recording and documenting transepithelial electrical resistance, permeability of fluorescence molecules to mimic drug delivery and growth kinetics quantified by MTT viability assay (to measure cell viability) and cell counts.

Laboratory Technologist

Jan. 2014 - Dec. 2015

Russel International Group of Schools, Benin City, Nigeria

- Maintaining safety and inventory for different STEM labs within the school as well as ordering supplies.
- Preparing samples and setting up different experiments for each lab at the recommendation of the subject head teacher.

Industry-based Internship

Feb. 2013 - June 2013

Oil Mill Analytical Laboratory, Okomu Oil Palm Company, Edo state, Nigeria

Supervisor: Engr. Omokaro Osaikhuiwuomwan

- Collection and determination of the free fatty acid in samples collected at different stages on the production of crude palm oil, palm kernel oil and palm kernel cake.